

Safety data sheet

Page: 1/12

BASF Safety data sheet
Date / Revised: 26.03.2025
Product: **Citral Extra**

Version: 2.0

(30035068/SDS_GEN_JP/EN)

Date of print: 25.02.2026

1. Substance/preparation and manufacturer/supplier identification

Product name:
Citral Extra

Use: Chemical, Chemical for detergents, Cosmetic and oral care chemical, flavoring substance

Manufacturer/supplier:

BASF Japan Ltd.
OVOL Nihonbashi Building 3F
3-4-4 Nihonbashi Muromachi, Chuo-ku
Tokyo, 103-0022, JAPAN
Telephone: +81-3-5290-3000
E-mail address: Japan-SDS-Info@basf.com

Emergency information:

Telephone: 03-6634-2245
+49 180 2273-112 (International emergency number)

2. Hazard identification

Classification of the substance and mixture:
Hazardous to the aquatic environment - acute: Cat.2
Skin sensitization: Cat.1
Acute toxicity: Cat.5 (dermal)
Acute toxicity: Cat.5 (oral)
Eye irritation: Cat.2A
Skin irritation: Cat.2

Label elements and precautionary statement:

Pictogram:



Signal Word:
Warning

Hazard Statement:

H319	Causes serious eye irritation.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H303 + H313	May be harmful if swallowed or in contact with skin.
H401	Toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves and eye protection or face protection.
P261	Avoid breathing mist or vapour or spray.
P273	Avoid release to the environment.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P302 + P312	IF ON SKIN: Call a POISON CENTER or a doctor/physician if you feel unwell.
P332 + P313	If skin irritation occurs: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Other hazards which do not result in classification:

When finely distributed on porous material, self-ignition is possible.

3. Composition/information on ingredients

Chemical nature

Substance nature: single product

3,7-dimethyl-2,6-octadien-1-al

CAS Number: 5392-40-5

ENCS: (2)-515

ISHL: (2)-515

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Hazardous ingredients

3,7-dimethyl-2,6-octadien-1-al

Content (W/W): > 90 % - <= 100 % Acute Tox.: Cat. 5 (oral)
CAS Number: 5392-40-5 Acute Tox.: Cat. 5 (dermal)
ENCS: (2)-515 Skin Irrit.: Cat. 2
ISHL: (2)-515 Eye Irrit.: Cat. 2A
Aquatic Acute: Cat. 2
Skin Sens.: Cat. 1

3-methylbut-2-en-1-ol

Content (W/W): > 0 % - < 0.2 % Flam. Liq.: Cat. 3
CAS Number: 556-82-1 Acute Tox.: Cat. 4 (oral)
ENCS: (2)-2393 Acute Tox.: Cat. 5 (dermal)
ISHL: (2)-2393 Skin Corr.: Cat. 1C
Eye Dam.: Cat. 1
Aquatic Acute: Cat. 3

Chemical Formula: C10 H16 O

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

dry powder, foam, carbon dioxide, water spray

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Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon oxides, harmful vapours
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Cool endangered containers with water-spray.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Information regarding personal protective measures, see section 8. Ensure adequate ventilation. Do not breathe vapour/spray. Avoid contact with the skin, eyes and clothing.

Environmental precautions:

Do not discharge into drains/surface waters/groundwater. Inform authorities in the event of product spillage to water courses or sewage systems.

Methods for cleaning up or taking up:

For small amounts: Pick up with suitable absorbent material. Do not use saw-dust or other combustible substances as an absorbant during cleanup.

For large amounts: Dike spillage. Pump off product.

Dispose of absorbed material in accordance with regulations. Mop up spills with non-flammable adsorbents (e.g. vermiculite, spill mats). Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner.

Additional information: When finely distributed on porous material, self-ignition is possible. Soiled textiles/cleaning rags made of natural fibres (e.g. of pure wool or of pure cotton) are capable of ignition and should not be used and/or must be disposed of in a safe manner.

7. Handling and Storage

Handling

Ensure thorough ventilation of stores and work areas. Wear suitable protective clothing and eye/face protection. Avoid contact with the skin, eyes and clothing. Keep container tightly sealed. This product may cause irritations; wash your hands after every contact.

Protection against fire and explosion:

Risk of self-ignition when a large surface area is produced due to fine dispersion. Soiled textiles / cleaning rags / adsorbents and Silica are capable of self ignition and should be wetted with water and must be disposed of in a safe manner. Avoid all sources of ignition: heat, sparks, open flame. Take precautionary measures against static discharges.

Storage

Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place. Protect from the effects of light.

8. Exposure controls and personal protection

Components with occupational exposure limits

3,7-dimethyl-2,6-octadien-1-al, 5392-40-5;

TWA value 5 ppm (ACGIHTLV), Inhalable fraction and vapor

Skin Designation (ACGIHTLV), Inhalable fraction and vapor

Danger of cutaneous absorption

Skin Designation (ACGIHTLV), Inhalable fraction and vapor

Danger of cutaneous absorption

Personal protective equipment

Respiratory protection:

Wear respiratory protection if ventilation is inadequate. Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

butyl rubber (butyl) - 0.7 mm coating thickness

fluoroelastomer (FKM) - 0.7 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Avoid contact with the skin, eyes and clothing. Do not breathe vapour/spray. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

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9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to yellowish	
Odour:	of lemon	
Odour threshold:	< 100 ppm	
pH value:	not applicable	
glass transition temperature:	-115 °C	(DSC (DIN 51007))
Boiling point:	approx. 230 °C (1,013 hPa) The substance / product decomposes.	(other)
Flash point:	98 °C Literature data.	(other)
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability (solid/gas):	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	225 °C Literature data.	(DIN 51794)
Thermal decomposition:	210 °C self-accelerating reaction	(DSC (DIN 51007))
Self ignition:	Based on its structural properties the product is not classified as self- igniting.	Test type: Spontaneous self- ignition at room-temperature.
SADT:	> 75 °C Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4)	
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	0.046 hPa (20 °C)	(calculated)

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	0.071 hPa (25 °C)	(calculated)
	1.003 hPa (59.29 °C)	(measured)
Density:	0.89 g/cm ³ (20 °C) Literature data.	
Relative density:	0.89 (20 °C) Literature data.	(other)
Relative vapour density (air):	5.24 (20 °C) Heavier than air.	(calculated)
Solubility in water:	moderately soluble 0.42 g/l (25 °C)	
Partitioning coefficient n-octanol/water (log Pow):	2.76 (25 °C)	(OECD Guideline 107)
Adsorption/water - soil:	log KOC: 2.1	(calculated)
Surface tension:	Based on chemical structure, surface activity is not to be expected.	
Viscosity, dynamic:	2.15 mPa.s (20 °C) 1.46 mPa.s (40 °C)	(calculated (from kinematic viscosity)) (calculated (from kinematic viscosity))
Viscosity, kinematic:	2.42 mm ² /s (20 °C) 1.67 mm ² /s (40 °C)	(OECD Guideline 114) (OECD Guideline 114)

10. Stability and Reactivity

Conditions to avoid:

Avoid all sources of ignition: heat, sparks, open flame. See SDS section 7 - Handling and storage.

Thermal decomposition: 210 °C (DSC (DIN 51007))
 self-accelerating reaction

Substances to avoid:
 acids, bases

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:

Self-ignition is possible when finely distributed on flammable surfaces in the presence of air.

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

Reactivity:

No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): approx. 6,800 mg/kg (BASF-Test)

Acute dermal toxicity

LD50 rat (dermal): > 2,000 mg/kg (BASF-Test)

Assessment of acute toxicity

Of low toxicity after single ingestion. Of low toxicity after short-term skin contact.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Irritation

Assessment of irritating effects:

Skin contact causes irritation. Eye contact causes irritation.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Irritant. (BASF-Test)

Serious eye damage/irritation rabbit: Irritant. (BASF-Test)

Respiratory/Skin sensitization

Assessment of sensitization:

Caused skin sensitization in animal studies. Caused sensitization in humans.

Experimental/calculated data:

Guinea pig maximization test guinea pig: skin sensitizing

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. In the majority of tests performed (mammalian cell culture) a mutagenic effect was not found. A mutagenic effect was also not observed in in-vivo assays.

Carcinogenicity

Assessment of carcinogenicity:

Results from a number of long-term carcinogenicity studies are available. Taking into account all of the information, there is no indication that the substance itself is carcinogenic.

Reproductive toxicity**Assessment of reproduction toxicity:**

The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity**Assessment of teratogenicity:**

Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)**Assessment of repeated dose toxicity:**

Prolonged repeated exposure caused inflammable degenerative processes in the respiratory tract of rats. Causes irritating effects at esophagus and the gastro-intestinal tract.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information**Ecotoxicity****Assessment of aquatic toxicity:**

Acutely toxic for aquatic organisms. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish:

LC50 (96 h) 6.8 mg/l, *Leuciscus idus* (DIN 38412 Part 15, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Aquatic invertebrates:

EC50 (48 h) approx. 7 mg/l, *Daphnia magna* (Directive 79/831/EEC, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Aquatic plants:

EC50 (72 h) 103.8 mg/l (growth rate), *Scenedesmus subspicatus* (DIN 38412 Part 9, static)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

Microorganisms/Effect on activated sludge:

EC50 (30 min) 2,100 mg/l, *Pseudomonas putida* (DIN 38412 Part 27 (draft), aquatic)

The product has low solubility in the test medium. An aqueous solution prepared with solubilizers has been tested. The details of the toxic effect relate to the nominal concentration.

EC20 (30 min) approx. 68 mg/l, activated sludge, domestic (OECD Guideline 209, aquatic)

Assessment of terrestrial toxicity:

Study scientifically not justified.

Mobility

Assessment transport between environmental compartments:

The substance will slowly evaporate into the atmosphere from the water surface.

Adsorption to solid soil phase is not expected.

Persistence and degradability

Elimination information:

92 % BOD of the ThOD (28 d) (OECD 301C; ISO 9408; 92/69/EWG, C.4-F) (aerobic, activated sludge, domestic)

> 90 % BOD of the ThOD (28 d) (OECD 301F; ISO 9408; 92/69/EWG, C.4-D) (aerobic, activated sludge, domestic)

Assessment of stability in water:

Substance is readily biodegradable, therefore hydrolysis is not expected to be relevant.

Study scientifically not justified.

Bioaccumulation potential

Assessment bioaccumulation potential:

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

13. Disposal Considerations

Observe national and local legal requirements.

14. Transport Information

Domestic transport:

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for	None known

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user

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

ERG-number: 171

See Section 15 for domestic transport regulations

15. Regulatory Information

Japanese Fire Service Law: 4th group, Petroleum Group 3, Not water soluble liquid

3,7-dimethyl-2,6-octadien-1-al

ISHL, SDS and Risk Assessment Requirements (JP)

MSDS, JP

3,7-dimethyl-2,6-octadien-1-al

ISHL, Hazardous Substances Labeling Requirements (JP)

ISHLLR, JP

Fire Service Law, Dangerous Substances (Japan)

Fire Service Law, Class 4, Flammable Liquids (JP)

03 2004

Storage Limit: 2000 Liters

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III
Class 3 Water-Insoluble Flammable Liquids

Skin and Eye Damaging or Absorbing Substances for PPE Requirement (ISHL Ministerial Ordinance Art. 594-2, MHLW Labour Standard Bureau Circular No. 0704-1), as amended (Japan)

| 3,7-dimethyl-2,6-octadien-1-al

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Compliant with JIS Z 7252/7253 :2019] Required items that are not described in this SDS indicate that there is no information

Some of the SDS and risk assessment required substances under Industrial Safety and Health Law are indicated with concentration range in SDS for trade secret protection.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.